

Performance Analytics

Kimia Farma - Big Data Analytics

Presented by

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SECTION 1

Profile Introduction & Courses/Certifications

Jovi Jethrovian Tampi

Second-year Global Supply Chain Management student at Sunway University Malaysia, with a strong foundation in data analytics, multimodal freight forwarding, and inventory control. Possess fundamentals in SQL, BigQuery, Google Looker Studio, and PowerBI, with practical and hands on academic project experience in supply chain integration, pharmaceutical inventory management, and freight forwarding case studies. Actively involved in Sunway Global Supply Chain Club and Sunway Indonesian Student Association (PPI).



<https://github.com/jovi-tampi/Big-Data-Analytics-I-Rakamin-Academy-x-Kimia-Farma/tree/main>



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Courses and Certification

UNESCAP CERTIFICATE IN MTO FREIGHT FORWARDING | SFFLA

October,

2025

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SECTION 2

Company and Project Background

BACKGROUND OF *kimia farma*

Kimia Farma is Indonesia's first and largest pharmaceutical company, established in 1817 and formally incorporated in 1971. Since February 2026, it operates as **Perusahaan Perseroan (Persero) PT Kimia Farma Tbk**, and is a subsidiary of PT Bio Farma (Persero) as part of Indonesia's state-owned Pharmaceutical Holding.



Kimia Farma Malioboro berdiri sejak 1865.



Cabang-cabang Kimia Farma (Tidak lengkap dikarenakan google maps tidak sanggup menunjukan semua)

The company runs an integrated healthcare business spanning **pharmaceutical manufacturing, R&D, distribution, and retail** — with more than 1,300 pharmacy outlets and 400 health clinics/laboratories across Indonesia, making it the market leader in Indonesian pharmaceutical retail.

Project Portfolio Background

Background

As part of the Rakamin Academy x Kimia Farma Big Data Analytics virtual internship, this project simulates the role of a Big Data Analytics Intern tasked with evaluating Kimia Farma's business performance from 2020 to 2023, using real transactional-scale data across the company's pharmacy and clinic network.

Available Data

- kf_final_transaction.csv ([link](#)),
- kf_inventory.csv ([link](#)),
- kf_kantor_cabang.csv ([link](#)),
- kf_product.csv ([link](#)).

kf_final_transaction (672,458 rows) — transaction-level sales records

kf_product (150 products) — product catalog and pricing

kf_kantor_cabang (1,725 branches) — branch location and customer ratings

kf_inventory (1,035,000 rows) — stock levels per branch/product

Problem Statement

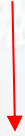
Kimia Farma's leadership needs a consolidated view of business performance across its 1,725 branches nationwide in order to answer: Which provinces and branches drive the most revenue and profit? Where do customer satisfaction gaps exist between branch reputation and actual transaction experience? And how has performance trended yoy from 2020–2023? This project gives insight, and transforms raw data to an interactive dashboard, to answer these questions and surface actionable business recommendations.

SECTION 3

Task Challenge (Phases 0–4)

Phase 0: Tools and Datasets Needed

TOOLS



Google
Big Query



Looker

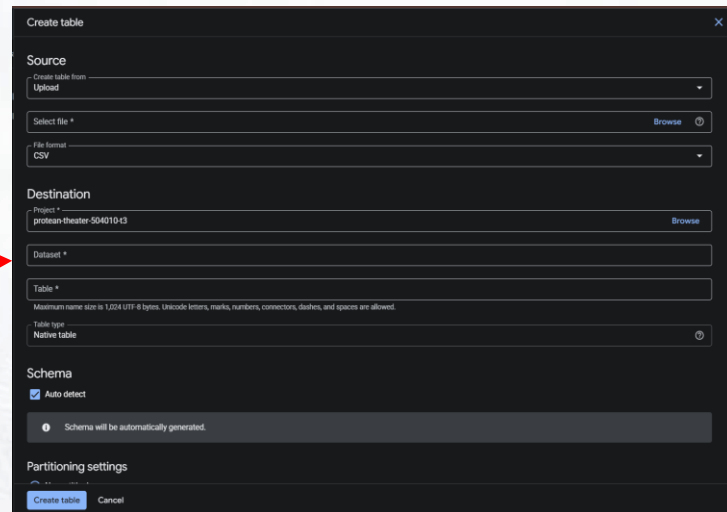
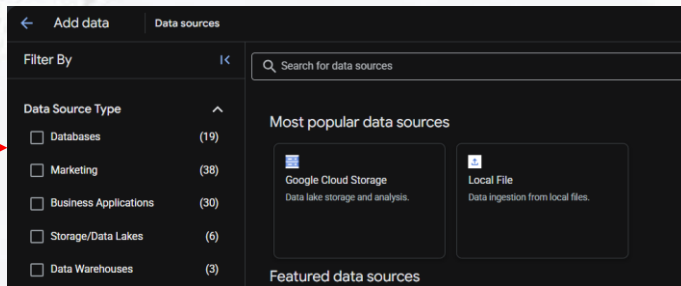
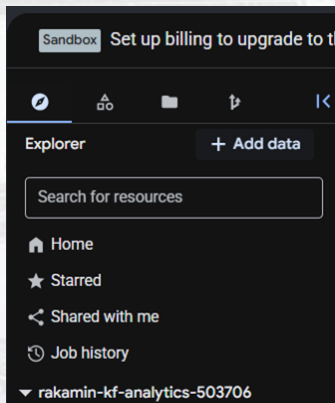
PART OF Google Cloud

DATASETS

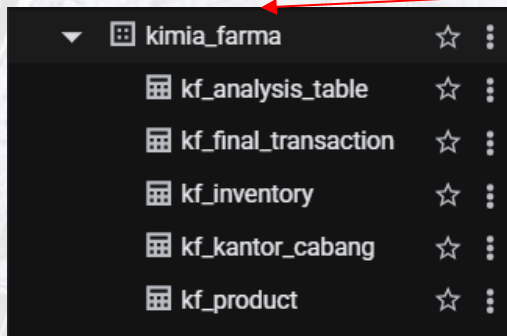


 kf_final_transaction	7/27/2026 2:40 PM	Microsoft Excel Co...	41,853 KB
 kf_inventory	7/27/2026 2:39 PM	Microsoft Excel Co...	85,916 KB
 kf_kantor_cabang	7/27/2026 2:39 PM	Microsoft Excel Co...	135 KB
 kf_product	7/27/2026 2:39 PM	Microsoft Excel Co...	12 KB

Phase 1: Importing Dataset to BigQuery



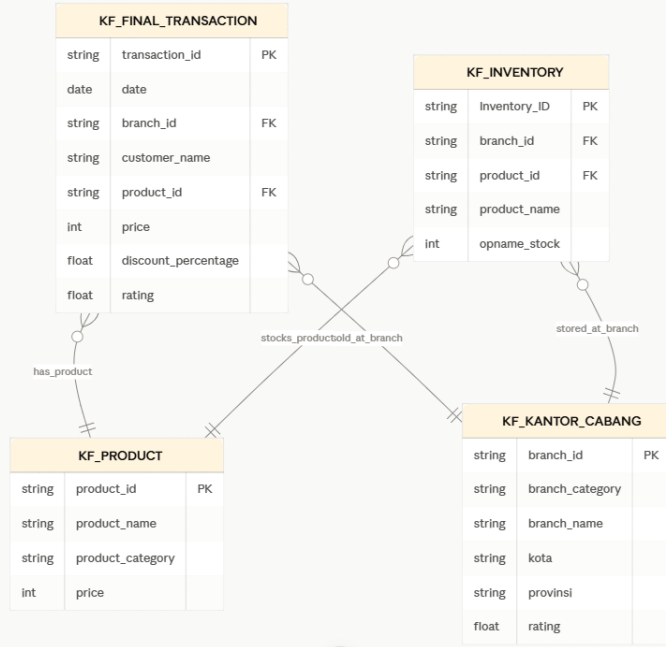
Created a dedicated GCP project (`Rakamin_KF_Analytics`) and BigQuery dataset (`kimia_farma`) to house all four source tables. Each CSV was imported using BigQuery's native schema auto detection, then validated to confirm correct data typing; most notably confirming the `date` column parsed correctly as `DATE` type in `M/D/YYYY` format (verified via null-count, min/max range, and monthly distribution checks) before proceeding to any downstream calculations.



Result: All 4 tables successfully imported - > means `kf_final_transaction` (672,458 rows), `kf_product` (150 rows), `kf_kantor_cabang` (1,725 rows), `kf_inventory` (1,035,000 rows). Zero data loss or parsing errors confirmed through validation queries.

Phase 2: Tabel Analisa

ERD



kf_analysis_table preview

Schema Details Preview Table Explorer Preview Insights Lineage Data Profile Data Quality

Filter Enter property name or value

Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags	Data Policies
transaction_id	STRING	NULLABLE	-	-	-	-	-	-	-
date	DATE	NULLABLE	-	-	-	-	-	-	-
branch_id	INTEGER	NULLABLE	-	-	-	-	-	-	-
branch_name	STRING	NULLABLE	-	-	-	-	-	-	-
kota	STRING	NULLABLE	-	-	-	-	-	-	-
provinsi	STRING	NULLABLE	-	-	-	-	-	-	-
rating_cabang	FLOAT	NULLABLE	-	-	-	-	-	-	-
customer_name	STRING	NULLABLE	-	-	-	-	-	-	-
product_id	STRING	NULLABLE	-	-	-	-	-	-	-
product_name	STRING	NULLABLE	-	-	-	-	-	-	-
actual_price	INTEGER	NULLABLE	-	-	-	-	-	-	-
discount_percentage	FLOAT	NULLABLE	-	-	-	-	-	-	-
percentage_gross laba	FLOAT	NULLABLE	-	-	-	-	-	-	-
nett_sales	FLOAT	NULLABLE	-	-	-	-	-	-	-
nett_profit	FLOAT	NULLABLE	-	-	-	-	-	-	-
rating_transaksi	FLOAT	NULLABLE	-	-	-	-	-	-	-

Edit schema View row access policies

Branch Granularity Validation

Row	unique_branch_ids	unique_branch_n...	unique_kota
1	1725	3	70

For the schema model, I chose star schema: `kf_final_transaction` sits as the fact table in the middle, with `kf_product` and `kf_kantor_cabang` as its direct dimension tables. `kf_inventory` isn't part of the required analysis table, but it shares the same two dimensions (`branch_id`, `product_id`), so it's drawn as a second, optional fact table off the same dims. (Only useful only if want to do stock level analysis).

Phase 2: Tabel Analisa Cont'd

PK/FK BREAKDOWN

Table	PK	FK
kf_final_transaction	transaction_id	product_id, branch_id
kf_product	product_id	—
kf_kantor_cabang	branch_id	—

Kf_inventory is excluded because it shares the same FKs but operates at a different grain (stock snapshot vs. transaction), so joining it would cause row fan out.

ACTUAL JOIN SYNTAX

```
FROM kf_final_transaction t
INNER JOIN kf_kantor_cabang b ON t.branch_id = b.branch_id
INNER JOIN kf_product p ON t.product_id = p.product_id
```

Because both relationships are many-to-one, INNER JOIN preserves all 672,458 rows with no duplication. This, confirms Part 1's granularity validation.

CARDINALITY STATEMENT

kf_product to kf_inventory:

Relationship: One-to-Many.

(A single product can have multiple inventory lines across multiple branches.)

kf_branch_office to kf_inventory:

Relationship: One-to-Many.

(A single branch can have multiple inventory lines for multiple products.)

kf_branch_office to kf_final_transaction:

Relationship: One-to-Many.

(A single branch can have multiple transactions, but each transaction only occurs at one branch.)

kf_product to kf_final_transaction:

Relationship: One-to-Many.

(A single product can appear in multiple transactions, but each transaction only references one product.)

Phase 3: BigQuery Syntax

Intro:

The analysis table was built using a two layer CTE (Common Table Expression) structure to separate column selection/renaming from the join logic, followed by a final SELECT that computes all derived business metrics.

FIRST CTE -> transaction_base prepare the fact table (& table creation wrapper)

```
1 CREATE OR REPLACE TABLE `rakamin-kf-analytics-503706.kimia_farma.kf_analysis_table` AS
2
3 WITH transaction_base AS (
4   SELECT
5     transaction_id,
6     date,
7     branch_id,
8     customer_name,
9     product_id,
10    price AS actual_price,
11    discount_percentage,
12    rating AS rating_transaksi
13  FROM `rakamin-kf-analytics-503706.kimia_farma.kf_final_transaction`
14 ),
```

CTE Purpose: Rename ambiguous columns early (price → actual_price, rating → rating_transaksi) before any joins happen – this avoids column-name collisions later, since kf_kantor_cabang also has a column called rating.

Wrapper Purpose: CREATE OR REPLACE allows safely re-running the script if the source data or logic changes, without manually dropping the table first.

SECOND CTE - joined: bring in the dimensions

```
16 joined AS (
17   SELECT
18     t.transaction_id,
19     t.date,
20     t.branch_id,
21     b.branch_name,
22     b.kota,
23     b.provinsi,
24     b.rating AS rating_cabang,
25     t.customer_name,
26     t.product_id,
27     p.product_name,
28     t.actual_price,
29     t.discount_percentage,
30     t.rating_transaksi
31  FROM transaction_base t
32  INNER JOIN `rakamin-kf-analytics-503706.kimia_farma.kf_kantor_cabang` b
33    ON t.branch_id = b.branch_id
34  INNER JOIN `rakamin-kf-analytics-503706.kimia_farma.kf_product` p
35    ON t.product_id = p.product_id
36 )
```

CTE #2 Purpose: Execute the two many-to-one joins established in the ERD (Slide 12), pulling in branch_name, kota, provinsi, rating_cabang from the branch dimension and product_name from the product dimension.

Phase 3: BigQuery Syntax Cont'd

FINAL SELECT -> Compute Derived Business Metrics

```
38 SELECT
39   transaction_id,
40   date,
41   branch_id,
42   branch_name,
43   kota,
44   provinsi,
45   rating_cabang,
46   customer_name,
47   product_id,
48   product_name,
49   actual_price,
50   discount_percentage,
51
52   CASE
53     WHEN actual_price <= 50000 THEN 0.10
54     WHEN actual_price > 50000 AND actual_price <= 100000 THEN 0.15
55     WHEN actual_price > 100000 AND actual_price <= 300000 THEN 0.20
56     WHEN actual_price > 300000 AND actual_price <= 500000 THEN 0.25
57     WHEN actual_price > 500000 THEN 0.30
58   END AS persentase_gross_labas,
59
60   ROUND(actual_price * (1 - discount_percentage), 2) AS nett_sales,
61
62   ROUND(
63     actual_price * (1 - discount_percentage) *
64     CASE
65       WHEN actual_price <= 50000 THEN 0.10
66       WHEN actual_price > 50000 AND actual_price <= 100000 THEN 0.15
67       WHEN actual_price > 100000 AND actual_price <= 300000 THEN 0.20
68       WHEN actual_price > 300000 AND actual_price <= 500000 THEN 0.25
69       WHEN actual_price > 500000 THEN 0.30
70     END, 2
71   ) AS nett_profit,
72
73   rating_transaksi
74
75 FROM joined;
```

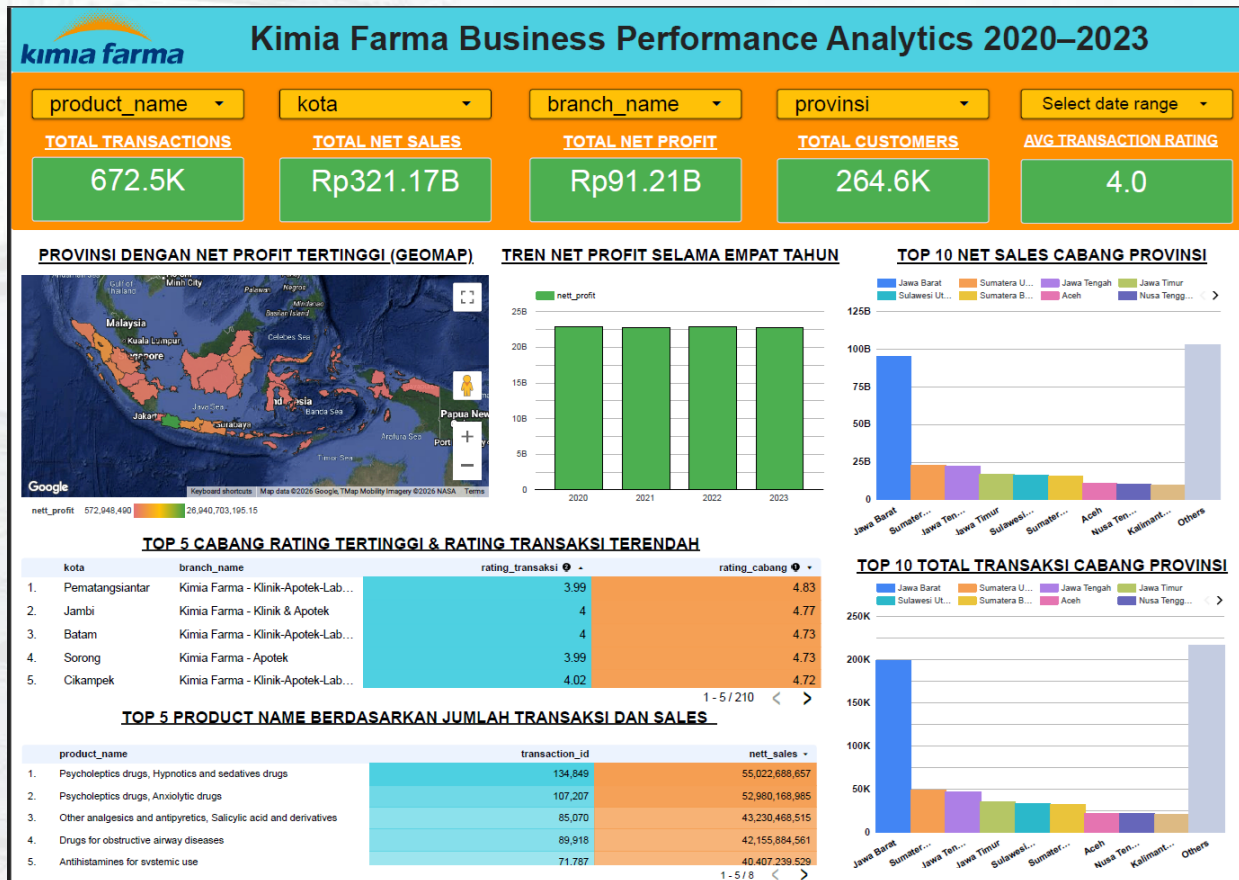
✓ Query completed

Using on-demand processing quota

Recent	Query results
▼ 8:00 PM	Job information Results Execution details Execution graph
✓ CREATE OR REPLA	ⓘ This statement replaced the table named kf_analysis_table.

Final Select Purpose: Implements the brief's tiered profit-margin rule directly as a **CASE** statement, then chains it into **nett_sales** (price after discount) and **nett_profit** ($\text{nett_sales} \times \text{margin rate}$).

Phase 4: Dashboard Performance Analytics



1. Header & Filters: Dashboard title, plus 5 filter controls (product_name, kota, branch_name, provinsi, date range) applying across all charts simultaneously.

2. Summary Scorecards: 5 KPIs: Total Transactions, Total Nett Sales, Total Nett Profit, Total Customers, Avg Transaction Rating.

3. Geo Map: Net profit distribution across Indonesian provinces.

4. YoY Trend: Net profit comparison across 2020–2023.

5. Top 10 Charts: Nett sales and total transactions ranked by province.

6. Rating Table: Top 5 branches by highest branch rating vs. lowest transaction rating.

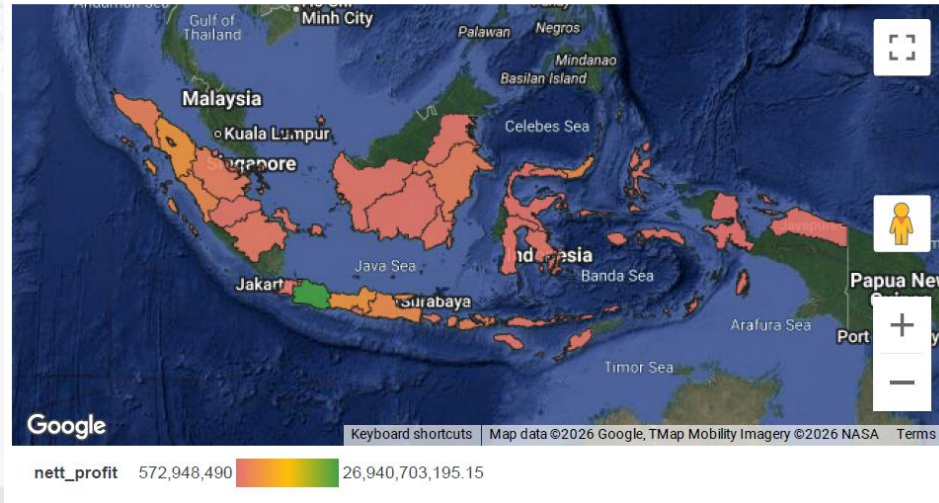
7. Bonus Chart: Top 5 products by transaction volume and sales.

SECTION 4

Analysis and Explanation

Drill Down Dashboard Analysis & Explanation

PROVINSI DENGAN NET PROFIT TERTINGGI (GEOMAP)



Analysis:

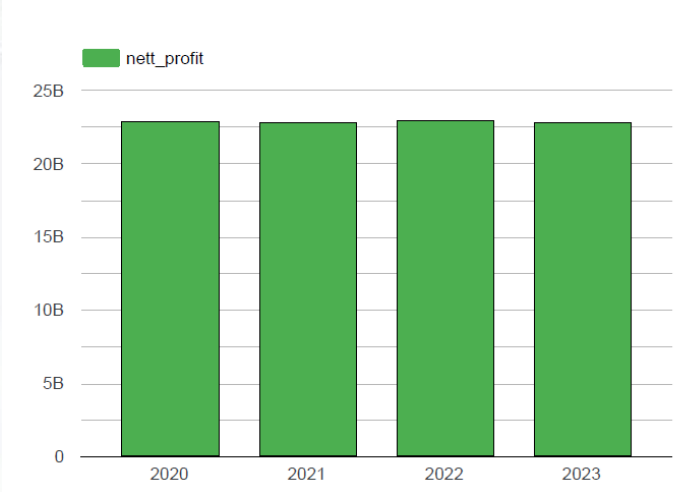
The geo map shows net profit concentration is heavily weighted toward Java based provinces, consistent with Indonesia's population and pharmacy network density. This aligns directly with the Top 10 charts (above), where Jawa Barat alone accounts for a disproportionately large share of both nett sales and transaction volume compared to any single other province.

Recommendation:

Given the concentration risk of relying heavily on Java-based provinces, Kimia Farma could evaluate whether under-performing provinces (visible as lighter-shaded regions on the map) suffer from fewer branch locations, lower foot traffic, or pricing/assortment mismatches → each requiring a different intervention.

Drill Down Dashboard Analysis & Explanation

TREN NET PROFIT SELAMA EMPAT TAHUN



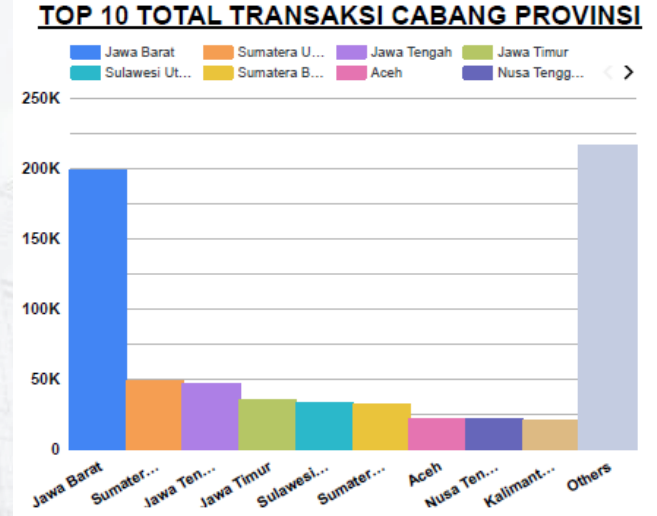
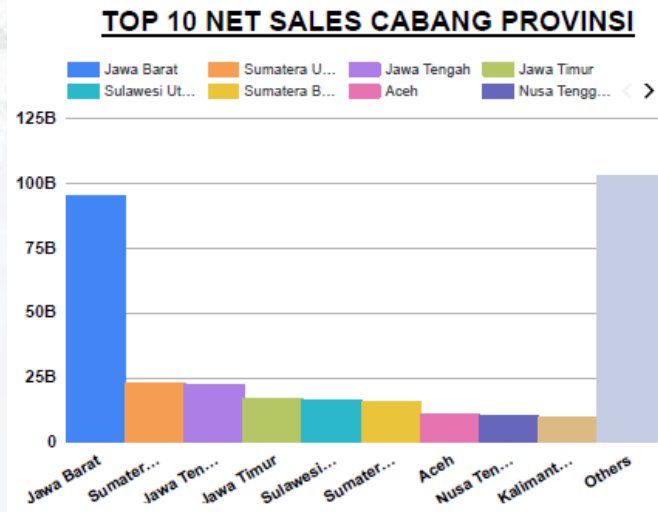
Analysis:

Net profit remained relatively stable across all four years, hovering in the 20 to 22B range with no dramatic growth or decline. This suggests consistent operational performance through the 2020–2023 window. Most notably including the COVID-19 period, where a pharmacy/healthcare business might have expected more volatility (either a spike from pandemic-related demand or disruption from mobility restrictions).

Recommendation:

Since profit has plateaued rather than grown, it's worth investigating whether this reflects a saturated market (branch count/coverage already maximized) or an opportunity for growth strategies, for example expanding into underserved provinces identified from the geo map, or increasing average transaction value.

Drill Down Dashboard Analysis & Explanation



Analysis:

Jawa Barat leads decisively in both metrics (100B in nett sales), roughly 4 times the next highest individual province. Both charts show a similar long-tail shape, which is a handful of provinces (Sumatera Utara, Jawa Tengah, Jawa Timur, Sulawesi Utara, Sumatera Barat) form a secondary tier, while the "Others" bucket (aggregating all remaining provinces) is nearly as large as Jawa Barat itself. This indicates performance is broadly distributed across many smaller provinces rather than concentrated in just a "top few."

Recommendation:

The fact that "Others" rivals the top province suggests meaningful revenue exists outside the headline performers -> worth a follow-up drill-down (filtering the dashboard by individual provinces within "Others") to identify any hidden high performers currently obscured by aggregation.

Drill Down Dashboard Analysis & Explanation

TOP 5 CABANG RATING TERTINGGI & RATING TRANSAKSI TERENDAH

	kota	branch_name	rating_transaksi	rating_cabang
1.	Pematangsiantar	Kimia Farma - Klinik-Apotek-Lab...	3.99	4.83
2.	Jambi	Kimia Farma - Klinik & Apotek	4	4.77
3.	Batam	Kimia Farma - Klinik-Apotek-Lab...	4	4.73
4.	Sorong	Kimia Farma - Apotek	3.99	4.73
5.	Cikampek	Kimia Farma - Klinik-Apotek-Lab...	4.02	4.72

1 - 5 / 210 < >

Analysis:

This table confirms a genuine and consistent pattern: all 5 branches shown have **rating_cabang** scores of 4.7 to 4.83, while their corresponding **rating_transaksi** sits notably lower at 3.99 to 4.02, which is roughly a 0.7 to 0.8 point gap. This means customers rate these branches highly as businesses/locations overall, but individual transaction experiences fall short of that reputation.

Recommendation:

This gap points to a specific, addressable service issue which is a likely transaction level friction (checkout speed, staff interaction, stock availability at point of sale) rather than a location/facility problem, since the branch itself is well-regarded. Kimia Farma could prioritize a customer service or checkout process audit at these 5 specific branches (Pematangsiantar, Jambi, Batam, Sorong, Cikampek) as a fast, targeted intervention.

Drill Down Dashboard Analysis & Explanation

TOP 5 PRODUCT NAME BERDASARKAN JUMLAH TRANSAKSI DAN SALES

	product_name	transaction_id	nett_sales ▾
1.	Psycholeptics drugs, Hypnotics and sedatives drugs	134,849	55,022,688,657
2.	Psycholeptics drugs, Anxiolytic drugs	107,207	52,980,168,985
3.	Other analgesics and antipyretics, Salicylic acid and derivatives	85,070	43,230,468,515
4.	Drugs for obstructive airway diseases	89,918	42,155,884,561
5.	Antihistamines for svstemic use	71.787	40.407.239.529
1 - 5 / 8 < >			

Analysis:

Psycholeptics (hypnotics/sedatives and anxiolytic drugs) rank #1 and #2 by both transaction count (134,849 and 107,207) and nett sales (Rp55B and Rp53B) which is meaningfully ahead of general analgesics, respiratory drugs, and antihistamines. This signals a notably high demand for mental health/sleep related medication across Kimia Farma's customer base.

Recommendation:

Given this demand concentration, ensuring consistent inventory availability for these specific product categories (cross-referencing with `kf_inventory` stock levels) is likely a high-value priority -> a stockout here would disproportionately affect both revenue and customer satisfaction given the volume involved.

LINKS

Github Here

Presentation Video Here

Thank You

